

TECHBIZ SECURITY

Ethical Hacking Internship Program

WEEK 2 – TASK 4 (EH)

LINUX FUNDAMENTALS AND KALI LINUX PRACTICAL

Submitted By:	MANOJ
Role:	Cyber Security Intern
Internship Organisation:	TechBiz Security
Internship ID:	TBS-INT-2026-518
Submitted Date:	27 August 2026
Task:	Week 2 – Task 4 (EH)

1. Introduction

This report documents the practical work completed under Week 2 – Task 4 (EH) of the Ethical Hacking Internship at TechBiz Security. The objective of this task was to build a working comfort level with Linux, since almost every activity in ethical hacking and cyber security – from reconnaissance and enumeration to exploitation and reporting – is carried out from a Linux-based command line. All practical work for this task was performed in a Kali Linux environment, which is the standard operating system used for penetration testing and security assessments.

The task covered basic Linux commands, file and directory operations, permission management using `chmod`, package installation and verification using APT (with Nmap as the example tool), and completion of the mandatory TryHackMe “Linux Fundamentals Part 1” room. Each activity was performed hands-on in the Kali terminal, and screenshots were captured as evidence of the commands executed and their output.

2. Task Objectives

- Understand the basic Kali Linux environment.
- Understand important Linux filesystem locations.
- Practice basic Linux commands.
- Perform file and directory operations.
- Understand Linux file permissions.
- Use `chmod` to modify permissions.
- Understand basic package management using APT.
- Install and verify Nmap.
- Complete the TryHackMe Linux Fundamentals Part 1 room.
- Document practical evidence through screenshots.

3. Kali Linux Environment

All practical work for this task was carried out inside a Kali Linux terminal. Kali Linux is a Debian-based distribution purpose-built for penetration testing, and getting comfortable with its command-line environment is a foundational skill for any ethical hacking role. The screenshot below shows basic identification and system-information commands executed in the terminal:

- **whoami** – confirmed the current logged-in user as *manuarya*.
- **pwd** – displayed the current working directory as */home/manuarya*.
- **ls** – listed the contents of the home directory.
- **uname -a** – displayed kernel and operating system information, confirming a Kali Linux system running kernel version 7.0.12-kali-amd64 on an x86_64 architecture.
- **id** – displayed the user ID, group ID, and group memberships for the current user, including membership in the *sudo* group.

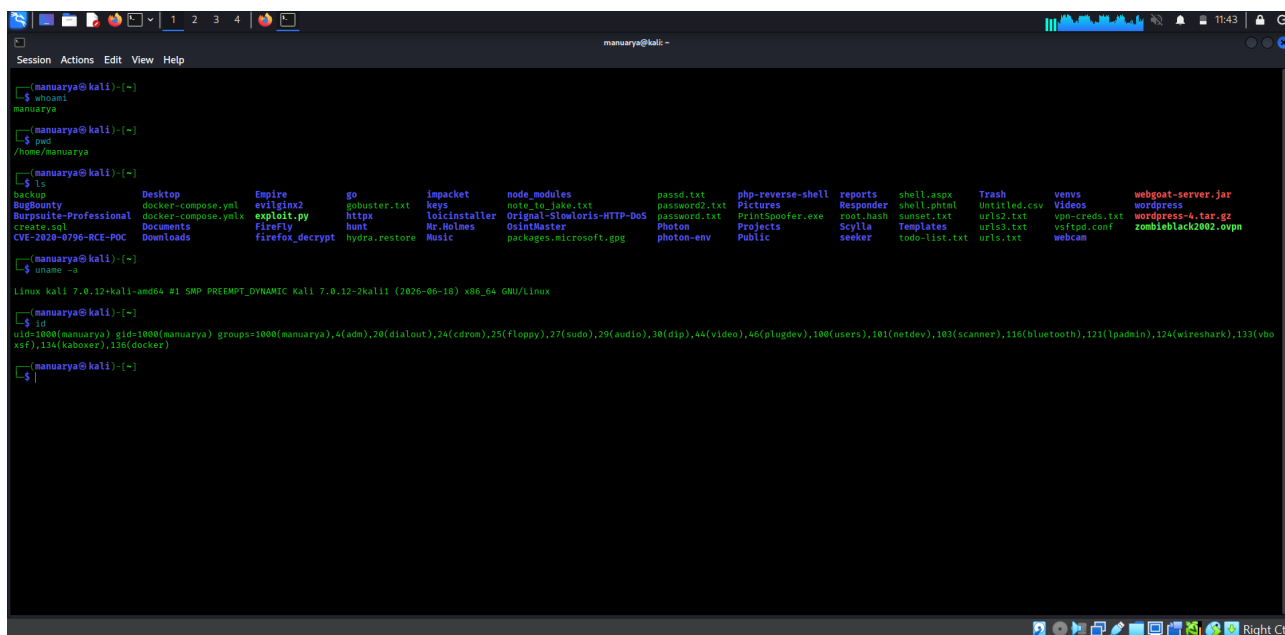
A screenshot of a Kali Linux terminal window. The terminal shows the user 'manuaria' at the prompt. They have run the commands 'whoami', 'pwd', and 'ls'. The 'ls' command output shows a list of files and directories in the home directory, including 'backup', 'Bugsounty', 'Burpsuite-Professional', 'CVE-2020-0796-RCE-POC', 'Desktop', 'docker-compose.yml', 'Documents', 'Downloads', 'Empire', 'evilsync2', 'exploit.py', 'FileFly', 'Firefox_decrypt', 'go', 'gobuster.txt', 'httpx', 'hunt', 'hydra.restore', 'Impacket', 'keys', 'loicinstaller', 'Mr.Holmes', 'Music', 'node_modules', 'net_tools', 'Original-Slowloris-HTTP-DoS', 'OsinMaster', 'packages.microsoft.gpg', 'passwd.txt', 'password2.txt', 'password.txt', 'Photon', 'photon-env', 'php-reverse-shell', 'Pictures', 'PrintSpoofer.exe', 'Projects', 'Public', 'reports', 'Responder', 'root.hash', 'Scylla', 'seeker', 'shell.aspx', 'shell.phtml', 'sunset.txt', 'Templates', 'todo-list.txt', 'Trash', 'untitled.csv', 'urls.txt', 'url5.txt', 'url6.txt', 'venvs', 'Videos', 'vpn-creds.txt', 'vsftpd.conf', 'webcam', 'webgoat-server.jar', 'wordpress', 'wordpress-4.tar.gz', and 'zombieblack2002.ovpn'. The terminal also shows the output of the 'uname -a' command, which displays system information like 'Linux kali 7.0.12-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 7.0.12-2kali1 (2026-06-18) x86_64 GNU/Linux'.

Figure 1 – Kali Linux Environment and Basic System Information

4. Linux Filesystem Overview

Before running commands, it is important to understand where key data lives on a Linux system. The table below summarises the directories covered as part of this task:

Directory	Description
/	Root of the Linux filesystem; every other directory branches from this location.
/home	Contains individual users' home directories and personal files, such as /home/manuaria.
/etc	Contains system-wide configuration files used by the operating system and installed services.
/var	Contains variable data such as logs, caches, and other files that change while the system runs.

Understanding these locations is directly useful during cybersecurity activities. During system enumeration, /home helps identify user accounts and personal files; /etc is a key location for reviewing configuration and identifying misconfigurations; and /var, particularly /var/log, is essential for log analysis and troubleshooting during an assessment.

5. Basic Linux Command Cheat Sheet

The table below summarises the core commands practiced during this task, along with their purpose. Commands marked as demonstrated were actually executed in the terminal as shown in the accompanying screenshots; the remaining commands were reviewed as part of the assigned Linux fundamentals but are included here for completeness of the cheat sheet.

Command	Purpose	Practical Usage
whoami	Displays the username of the currently logged-in user.	Demonstrated in Figure 1.
pwd	Prints the full path of the current working directory.	Demonstrated in Figures 1, 2 and 3.
ls	Lists the files and directories inside the current or specified directory.	Demonstrated in Figure 1; ls -l used in Figure 3 to view permissions.

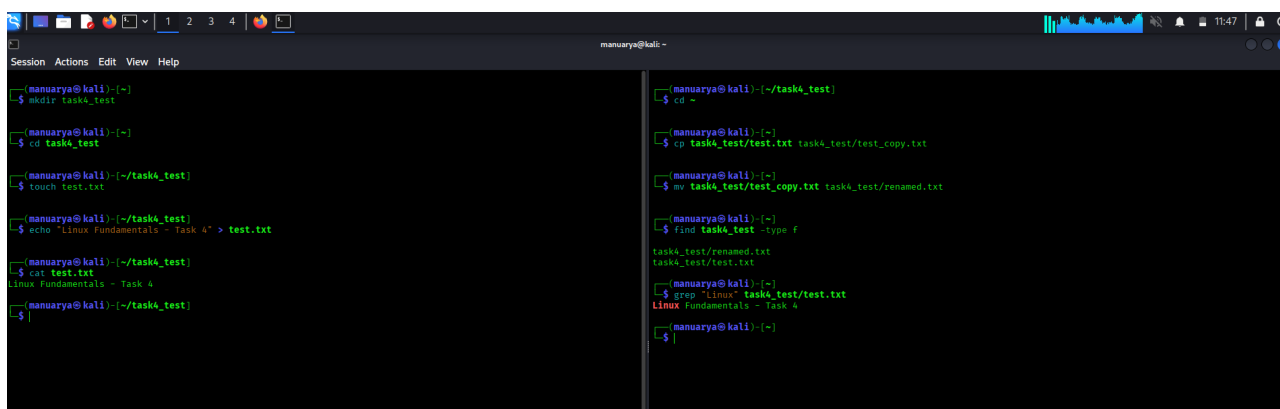
Command	Purpose	Practical Usage
cd	Changes the current working directory to the specified path.	Demonstrated in Figures 2 and 3.
cat	Displays the full contents of a text file on the terminal.	Demonstrated in Figures 2 and 3 to read test.txt.
mkdir	Creates a new, empty directory.	Demonstrated in Figures 2 and 3 (task4_test, task4_lab).
rm	Deletes files or directories.	Reviewed as part of task fundamentals; not directly demonstrated in the screenshots.
cp	Copies a file or directory to a new location or name.	Demonstrated in Figure 2 (test.txt copied to test_copy.txt).
mv	Moves or renames a file or directory.	Demonstrated in Figure 2 (test_copy.txt renamed to renamed.txt).
chmod	Changes the read, write and execute permissions of a file or directory.	Demonstrated in Figure 3 (chmod 755 applied to test.txt).
chown	Changes the owner and/or group of a file or directory.	Reviewed as part of task fundamentals; not directly demonstrated in the screenshots.
sudo	Executes a command with elevated (root) privileges.	Used implicitly for package installation; role explained in Section 8.
apt	Debian/Kali package manager used to install, update and remove software packages.	Used to install Nmap; verified in Figure 4.
uname	Displays system and kernel information; -a shows all available details.	Demonstrated in Figure 1 (uname -a).
id	Displays the user ID, group ID and group memberships of the current user.	Demonstrated in Figure 1.
touch	Creates a new, empty file, or updates the timestamp of an existing file.	Demonstrated in Figures 2 and 3 (test.txt).
echo	Prints text to the terminal; combined with > to write text into a file.	Demonstrated in Figures 2 and 3 to write text into test.txt.
find	Searches for files and directories matching a given name or type.	Demonstrated in Figure 2 (find task4_test -type f).
grep	Searches for a text pattern inside one or more files.	Demonstrated in Figure 2 and in the TryHackMe exercise (Figure 5).
head	Displays the first lines of a file (default: first 10 lines).	Demonstrated in Figure 3 (head -n 1).
tail	Displays the last lines of a file (default: last 10 lines).	Demonstrated in Figure 3 (tail -n 1).

6. Practical File and Directory Operations

Figure 2 shows the practical workflow followed for file and directory operations. The following steps were carried out in sequence in the Kali terminal:

- Created a new directory named **task4_test** using **mkdir**.
- Entered the directory using **cd task4_test**.
- Created an empty file named **test.txt** using **touch**.
- Wrote the text “Linux Fundamentals - Task 4” into the file using **echo** with output redirection (>).
- Read the contents of the file back using **cat test.txt**, confirming the text was saved correctly.
- Copied the file to a new file, **test_copy.txt**, using **cp**.
- Renamed the copied file to **renamed.txt** using **mv**.
- Used **find task4_test -type f** to list all regular files inside the task4_test directory, which returned renamed.txt and test.txt.
- Used **grep "Linux" task4_test/test.txt** to search for the word “Linux” inside the file, which successfully matched the line written earlier.

These operations reflect tasks that are performed regularly during cybersecurity work: navigating an unfamiliar filesystem, creating working directories to organise evidence, duplicating files before modifying them, and using find and grep to quickly locate files or search for specific strings within logs and configuration data.



```
(manuaria@kali)~$ mkdir task4_test
(manuaria@kali)~$ cd task4_test
(manuaria@kali)~/task4_test$ touch test.txt
(manuaria@kali)~/task4_test$ echo "Linux Fundamentals - Task 4" > test.txt
(manuaria@kali)~/task4_test$ cat test.txt
Linux Fundamentals - Task 4
(manuaria@kali)~/task4_test$ cp task4_test/test.txt task4_test/test_copy.txt
(manuaria@kali)~/task4_test$ mv task4_test/test_copy.txt task4_test/renamed.txt
(manuaria@kali)~/task4_test$ find task4_test -type f
task4_test/renamed.txt
task4_test/test.txt
(manuaria@kali)~/task4_test$ grep "Linux" task4_test/test.txt
Linux Fundamentals - Task 4
(manuaria@kali)~/task4_test$
```

Figure 2 – Linux File and Directory Operations

7. File Permissions and chmod

Every file and directory on a Linux system has permissions assigned to three categories of users: the **Owner** of the file, the **Group** associated with the file, and **Others** (everyone else on the system). Each category can be granted a combination of three permission types: **r** (read), **w** (write), and **x** (execute).

Figure 3 demonstrates a practical permission change using chmod. The permission string on test.txt was observed before and after the change:

State	Permission String	Meaning
Before chmod 755	-rw-rw-r--	Owner: read/write, Group: read/write, Others: read only.
After chmod 755	-rwxr-xr-x	Owner: read/write/execute, Group: read/execute, Others: read/execute.

The chmod 755 command sets the following permissions:

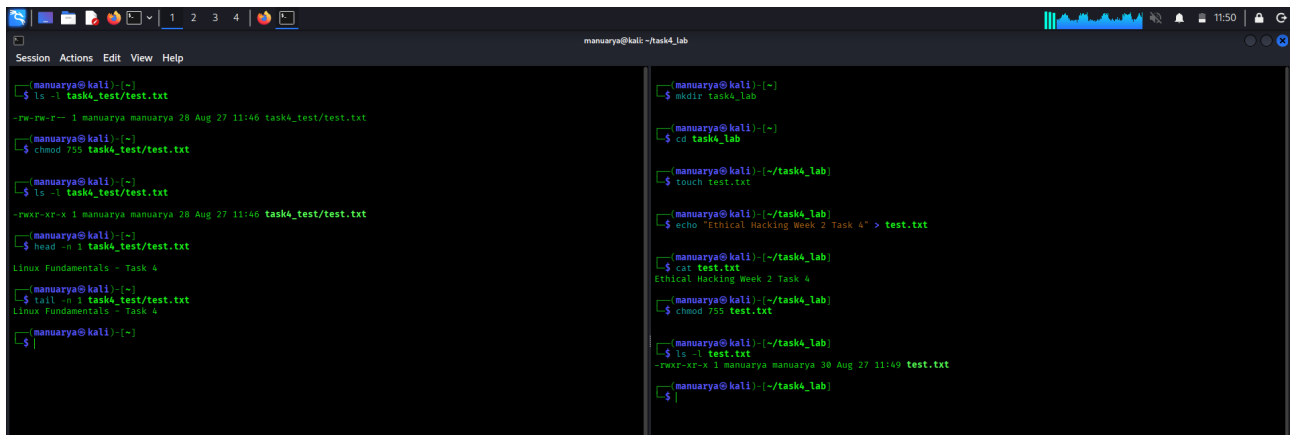
- **Owner:** read + write + execute
- **Group:** read + execute

- **Others:** read + execute

In numeric form, each digit is the sum of the permission values (read = 4, write = 2, execute = 1):

- **7** = 4 (read) + 2 (write) + 1 (execute) – applied to the Owner.
- **5** = 4 (read) + 1 (execute) – applied to the Group.
- **5** = 4 (read) + 1 (execute) – applied to Others.

After applying `chmod 755`, the **head -n 1** and **tail -n 1** commands were used to confirm that the file content – “Linux Fundamentals - Task 4” – remained unchanged after the permission modification. Understanding and correctly setting file permissions is an important security practice, since overly permissive files (for example, world-writable scripts or configuration files) are a common source of privilege-escalation vulnerabilities that ethical hackers are trained to identify.



```
manuaria@kali: ~$ ls -l task4_test/test.txt
-rw-rw-r-- 1 manuaria manuaria 28 Aug 27 11:46 task4_test/test.txt

manuaria@kali: ~$ chmod 755 task4_test/test.txt

manuaria@kali: ~$ ls -l task4_test/test.txt
-rwxr-xr-x 1 manuaria manuaria 28 Aug 27 11:46 task4_test/test.txt

manuaria@kali: ~$ head -n 1 task4_test/test.txt
Linux Fundamentals - Task 4

manuaria@kali: ~$ tail -n 1 task4_test/test.txt
Linux Fundamentals - Task 4

manuaria@kali: ~$

manuaria@kali: ~$ mkdir task4_lab

manuaria@kali: ~$ cd task4_lab

manuaria@kali: ~/task4_lab$ touch test.txt

manuaria@kali: ~/task4_lab$ echo "Ethical Hacking Week 2 Task 4" > test.txt

manuaria@kali: ~/task4_lab$ cat test.txt
Ethical Hacking Week 2 Task 4

manuaria@kali: ~/task4_lab$ chmod 755 test.txt

manuaria@kali: ~/task4_lab$ ls -l test.txt
-rwxr-xr-x 1 manuaria manuaria 30 Aug 27 11:49 test.txt

manuaria@kali: ~/task4_lab$
```

Figure 3 – Changing File Permissions Using `chmod 755`

8. Package Management and Nmap Installation

Kali Linux, being Debian-based, uses APT (Advanced Package Tool) for installing, updating and removing software packages. APT resolves dependencies automatically and pulls packages from configured repositories, which makes it the standard way to install security tools on Kali.

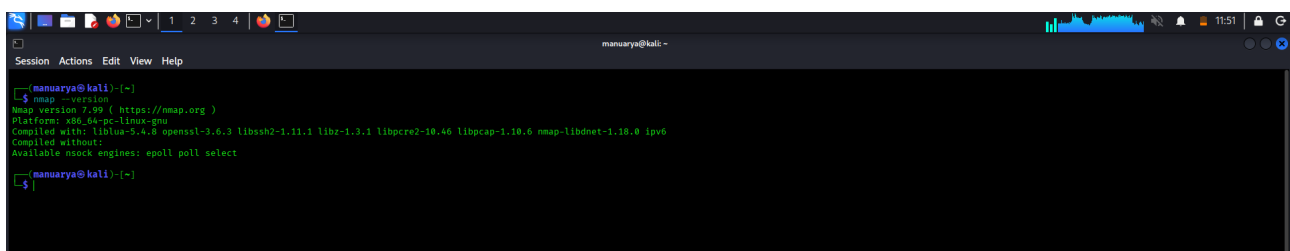
The installation command used for this task follows the pattern below:

```
sudo apt install nmap
```

Each component of this command has a specific role:

- **sudo** – executes the command with elevated (root) privileges, which are required to install software.
- **apt** – the package management utility used on Debian-based systems such as Kali Linux.
- **install** – tells APT to install the specified package.
- **nmap** – the package being installed, in this case the Nmap network scanning tool.

After installation, the presence and version of Nmap were verified using **nmap --version**. As shown in Figure 4, the command confirmed that Nmap version 7.99 is installed and available on the system, along with details of the libraries it was compiled with.



```
manuaria@kali:~$ nmap --version
nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.0 openssl-3.0.3 libssh2-1.11.1 libz-1.3.1 libpcre2-10.46 libpcap-1.10.6 nmap-libdnet-1.10.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
manuaria@kali:~$
```

Figure 4 – Nmap Installation Verification

9. TryHackMe – Linux Fundamentals Part 1

As part of the mandatory practical requirement for this task, the TryHackMe room “Linux Fundamentals (Pt1)” was completed. The room reinforced the Linux command-line fundamentals practiced locally in Kali by applying them inside a live, browser-based Linux terminal.

In the final practical exercise of the room, the following commands were used to navigate to the home directory, locate a log file, and search its contents for a hidden flag:

- **pwd** – confirmed the working directory as /home/tryhackme.
- **ls** – listed the contents of the home directory, including access.log and folder1 to folder4.
- **cd folder1** – changed into folder1, which contained access.log and passwords.txt.
- **cat passwords.txt** – displayed the contents of passwords.txt.
- **grep THM access.log** – searched access.log for the string “THM”, returning a log entry containing the flag.

The grep command successfully located the flag within the log file, and the retrieved flag was submitted as the answer to the room's question:

```
THM{ACCESS}
```

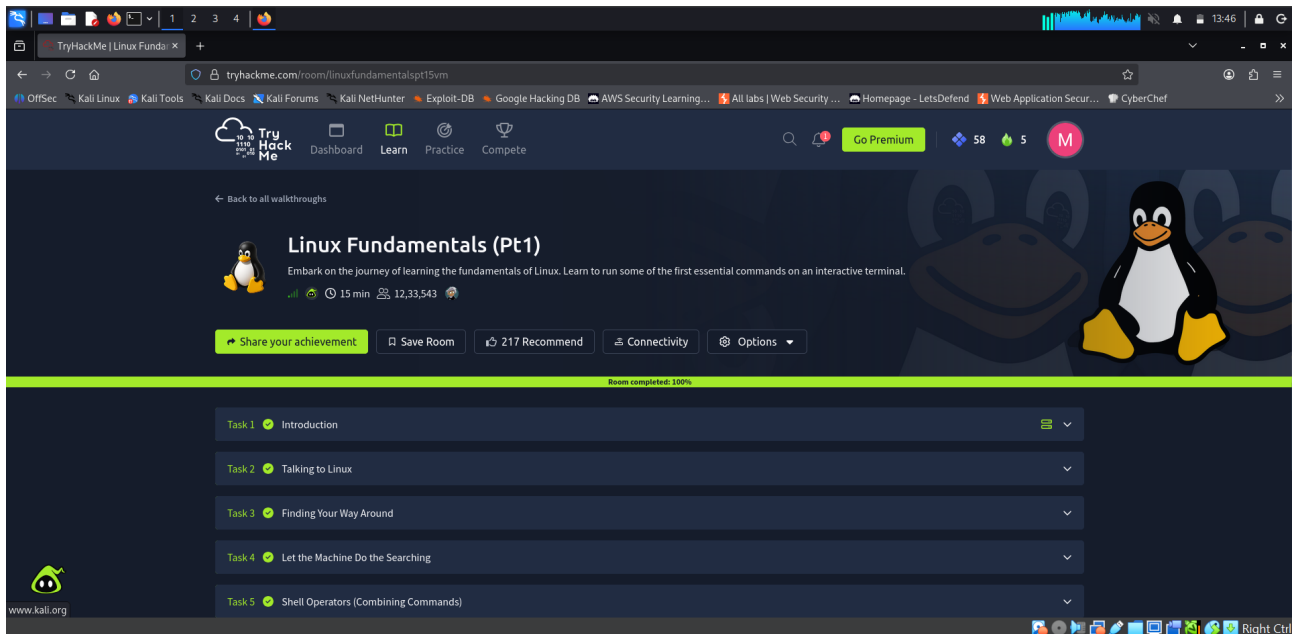


Figure 5 – TryHackMe Linux Fundamentals Practical Exercise

10. TryHackMe Completion Evidence

Figure 6 shows the TryHackMe “Linux Fundamentals (Pt1)” room dashboard with a completion status of 100%, confirming that all tasks in the room, including the grep exercise described above, were completed.

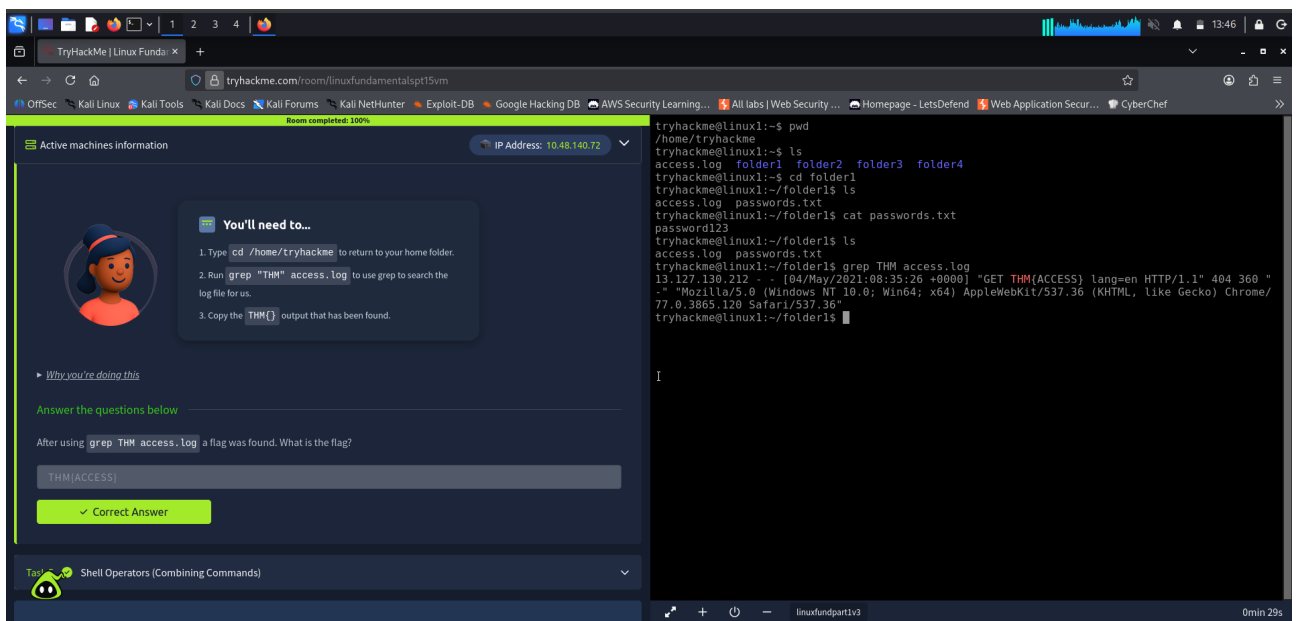


Figure 6 – TryHackMe Linux Fundamentals Part 1 Completion Evidence

11. Practical Learning Outcomes

The following skills were practically demonstrated during this task:

- Linux terminal navigation.
- Identifying the current user and working directory.
- Listing and inspecting files.
- Creating directories and files.
- Writing and reading file content.
- Copying and moving files.
- Searching files and file contents.
- Understanding Linux permissions.
- Modifying permissions with `chmod 755`.
- Understanding basic APT package management.
- Installing and verifying Nmap.
- Using `grep` during a TryHackMe Linux exercise.

These skills form a practical foundation for ethical hacking work. Comfortable Linux navigation and file handling directly support vulnerability assessment and penetration testing engagements, where most tools run from the command line. They are equally relevant to SOC operations and security analysis, where analysts regularly navigate log directories, search files for indicators of compromise, and manage permissions on evidence and scripts.

12. Cybersecurity Relevance

The Linux fundamentals practiced in this task are directly applicable to authorised ethical hacking and security work:

- Linux command-line skills support efficient system enumeration during an assessment.
- File navigation helps locate configuration files, logs, scripts, and evidence on a target system.
- `grep` and `find` help search large amounts of system information quickly and accurately.
- `chmod` and permission awareness help identify and understand access-control issues.
- `sudo` knowledge helps understand privilege boundaries between standard and elevated users.
- APT knowledge helps install and manage security tools on Debian-based systems such as Kali.
- Nmap is a fundamental network scanning and enumeration tool used in authorised security assessments to identify live hosts, open ports and running services.

13. Evidence Summary

Evidence	Activity Demonstrated	Screenshot
1	Basic Linux commands and system information	Figure 1
2	File and directory operations	Figure 2
3	<code>chmod 755</code> and permission verification	Figure 3
4	Nmap installation verification	Figure 4
5	TryHackMe Linux command-line exercise	Figure 5
6	TryHackMe room completion	Figure 6

14. Conclusion

Week 2 – Task 4 provided practical exposure to Kali Linux and the essential Linux command-line operations that underpin ethical hacking work. Hands-on practice with file and directory management, permission control using `chmod`, package management through APT, and installation and verification of Nmap strengthened the working knowledge required for future security assessments. Completion of the TryHackMe Linux Fundamentals Part 1 room further reinforced these skills in an independent, scenario-based environment. Overall, this task built a solid Linux foundation that will support the more advanced enumeration, exploitation, and reporting tasks planned for later stages of the internship.